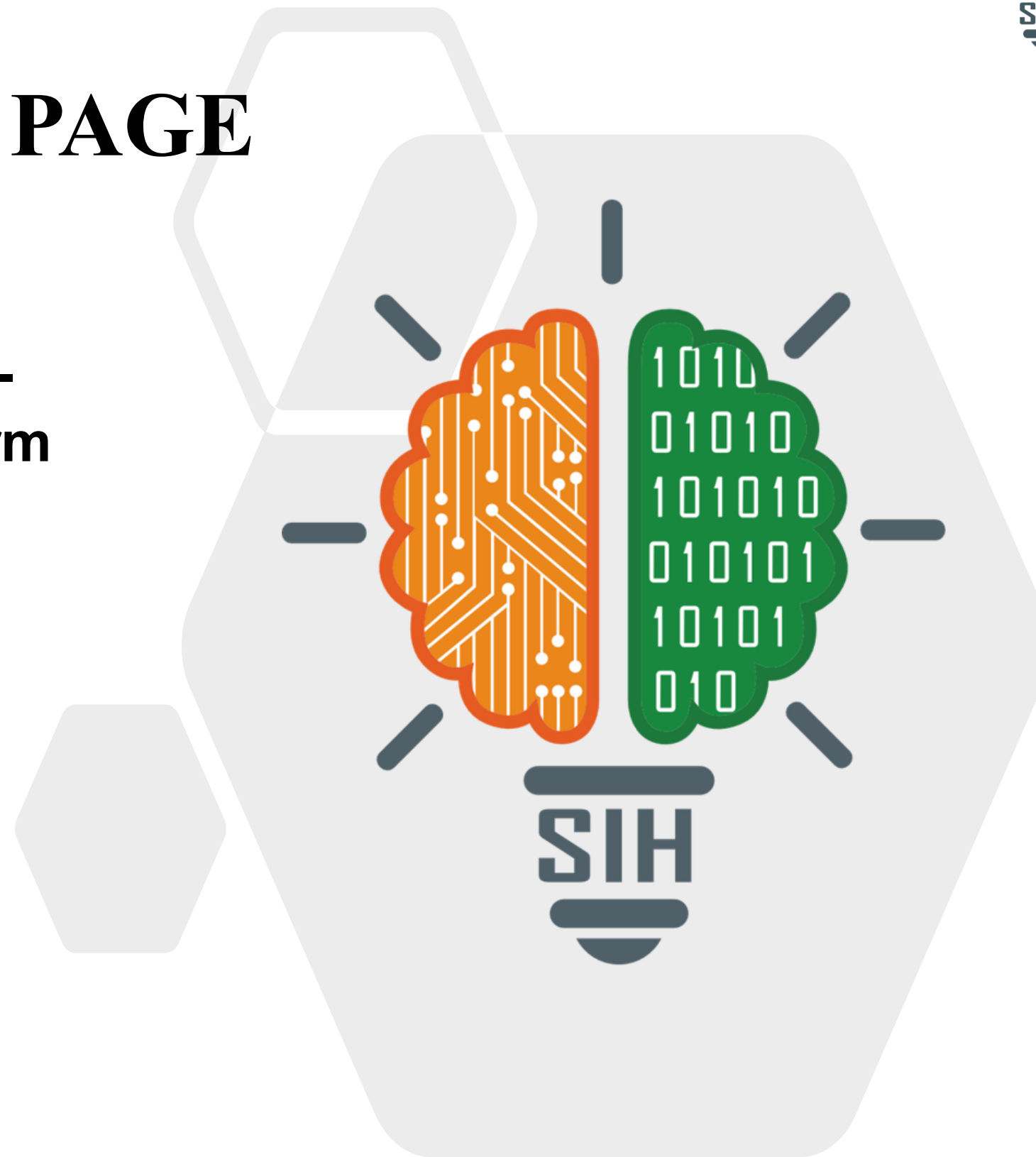


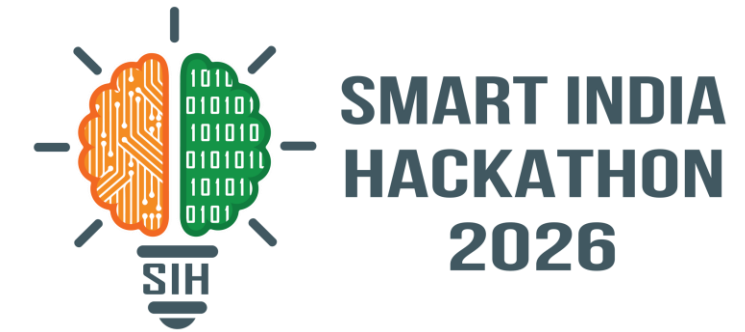
TITLE PAGE

- Problem Statement ID - SIH26067
- Problem Statement Title - Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.
- Theme - Disaster Management
- PS Category - Software
- Team ID - K26125
- Team Name - Team Null



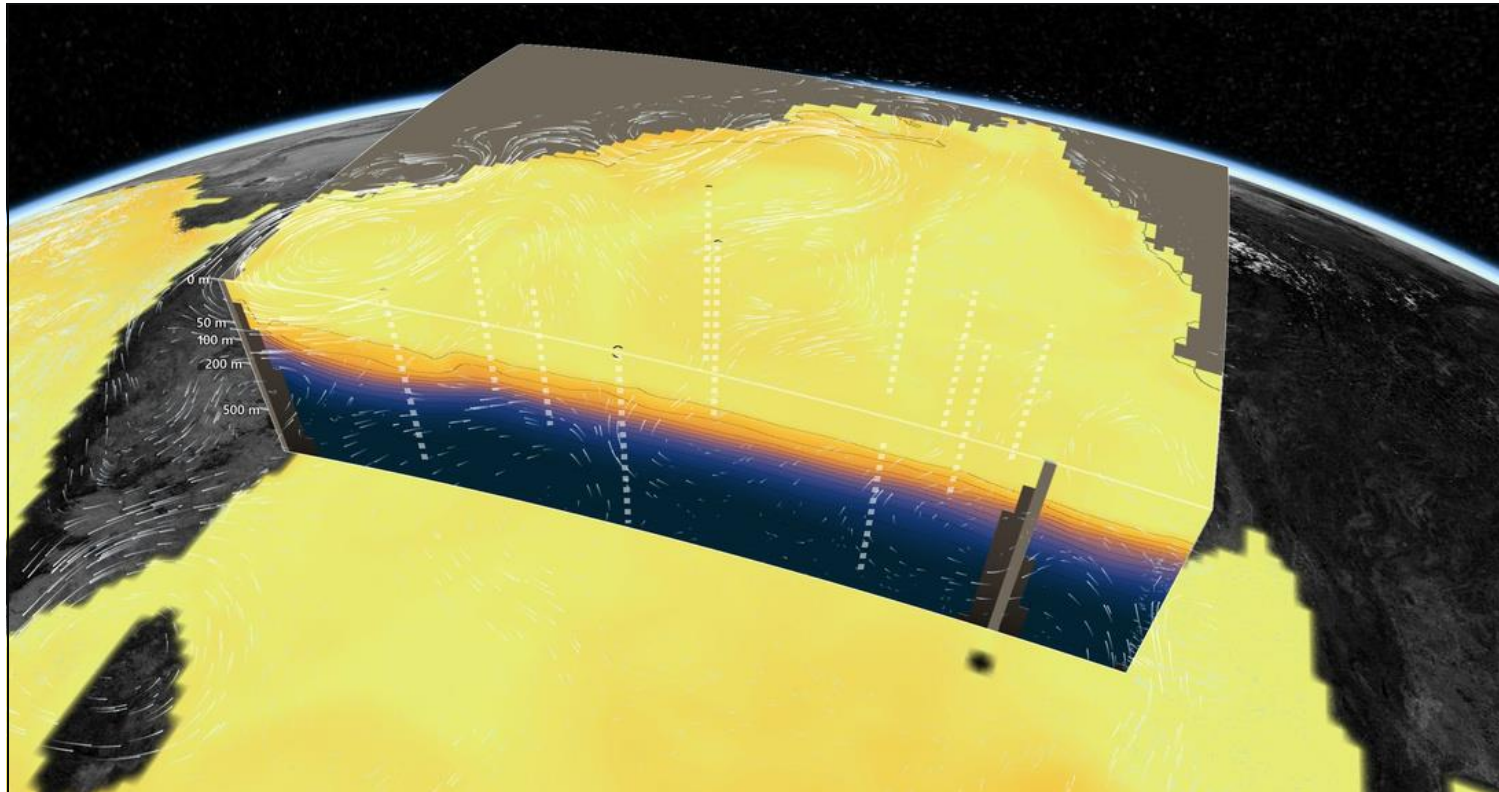


VWater: Ocean in 3D



PROPOSED SOLUTION

A browser platform that cuts a **3D block out of the ocean model, anywhere, on any day from 1993 to the forecast**, and stands the real Argo floats inside it.



Working prototype, captured. The Bay of Bengal on 2020-05-14, before Cyclone Amphan, cut open to 1,000 m on the model's 35 native depth levels; each dashed stick is one of 17 Argo floats within two days.

DETAILED EXPLANATION

Model: Copernicus GLORYS12 and PISCES, read from the cloud; the INCOIS Argo analysis for the Bay

Instruments: Argo and BGC floats, gliders, RAMA moorings, CTD casts, each with its QC state and source file

Views: Region 3D, Map 2D, Globe, Fly, immersive; winds, waves, currents, fishing zones

HOW IT ADDRESSES THE PROBLEM

The five gaps the problem statement lists, and our answer to each.

No browser-based 3D, depth-resolved view of model fields
a block of any box and day, cut open, in any browser; the INCOIS analysis as a true 3D volume

Argo and glider profiles not shown alongside model fields
floats stand inside the model as sticks; click one for its profile against the model

No controls for variable, depth slice, time step, colour bar
23 variables, cut planes, time-step animation (Play), palette, range and log scale

New data streams need re-engineering
a variable or ML product is one catalogue line; a sensor, one registry entry

No tools for rapid understanding and decisions
an AI assistant builds the view you ask for; fishing advisories; cyclone heat potential

Also asked for by name, and built

Isosurface (the 20 °C isotherm, Bay volume) · layer opacity · vertical exaggeration · NetCDF and text parsers · REST API · OGC WMS (Bay layers) · ADCP and HF-radar readers

INNOVATION AND UNIQUENESS

Floats inside the model

click one: its readings beside the model's, on the float's own day

No archive to build

reads only the chunks a box needs from Copernicus's cloud stores

The model is checked too

impossible values masked and counted; INCOIS error drawn faint

An assistant that acts (AI Chatbot)

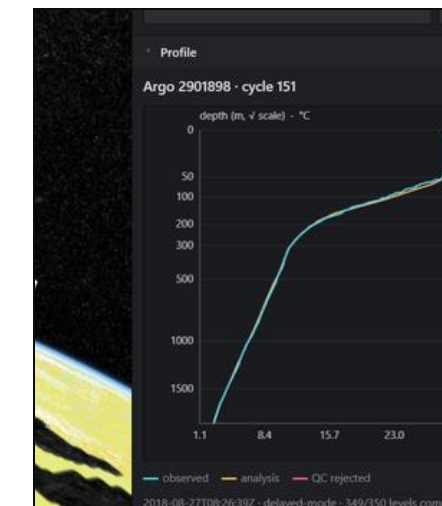
data only via our API tools; untraceable numbers flagged; built the asked-for cube 5 of 5 times, median 4.6 s

A course on live data

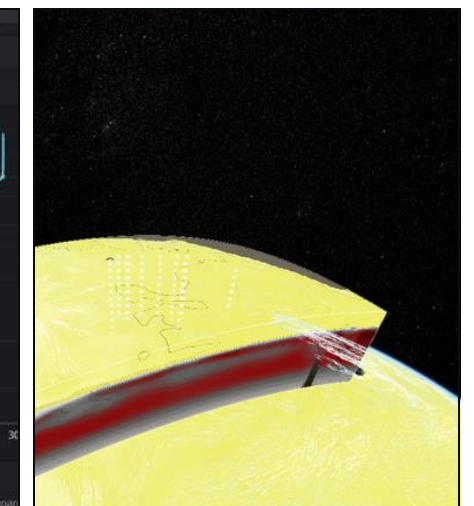
six lessons, class 8–12; quiz answers read from the cube on screen

A flight view (surface through the sky)

fly a few kilometres above the sea and coast, as a pilot sees it



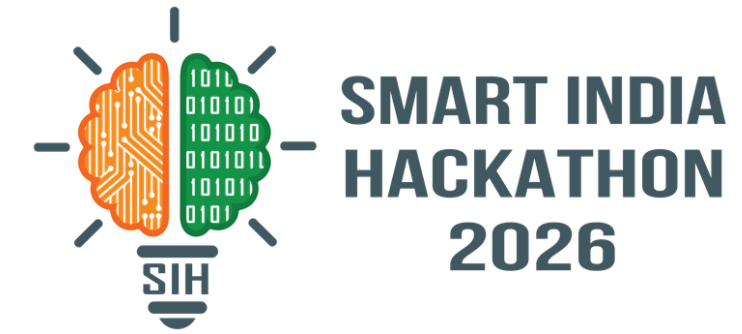
a float's profile vs model



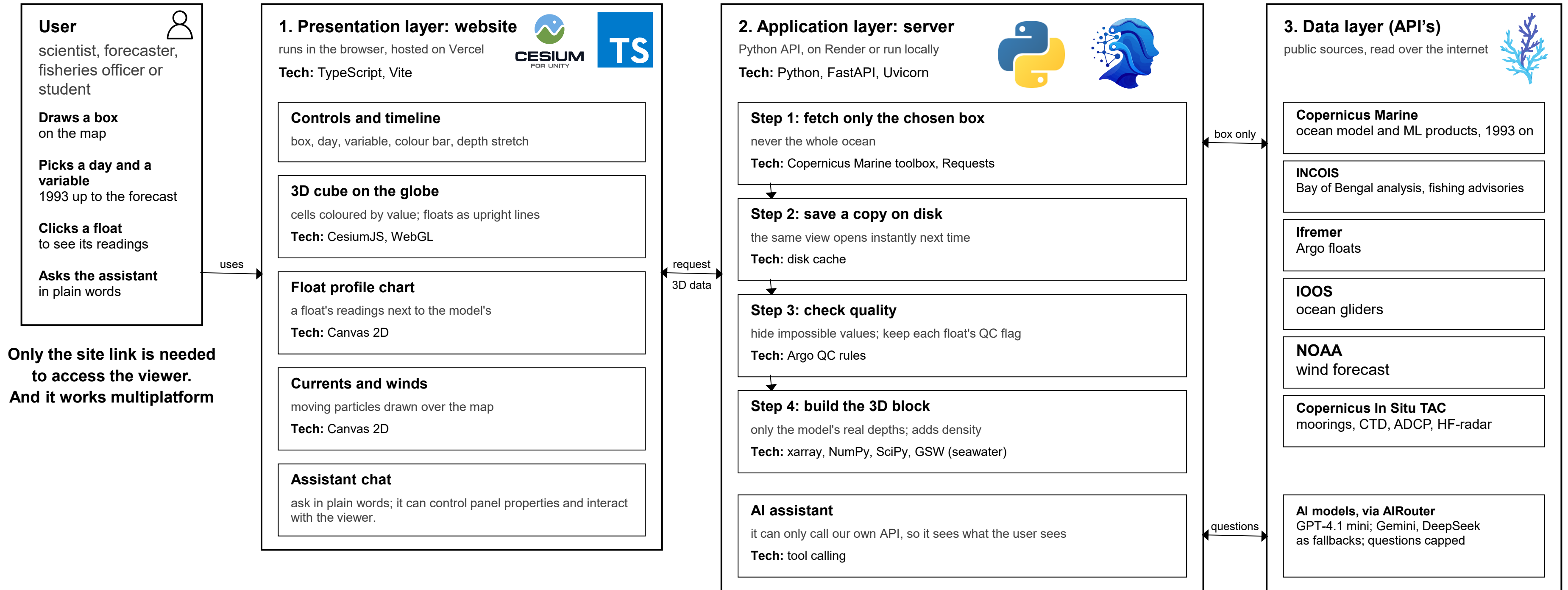
Arabian Sea, built by the assistant



TECHNICAL APPROACH



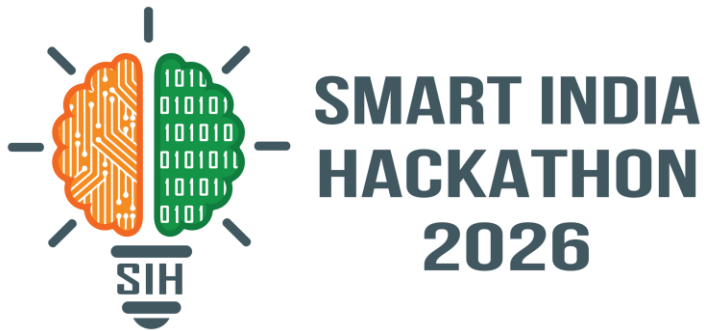
Software architecture: three layers, end to end



What is calculated at each position: Longitude, latitude → where it sits on the globe Depth → how far down (can be stretched) Value → its colour on the colour bar



FEASIBILITY AND VIABILITY



ANALYSIS OF FEASIBILITY: MEASURED BY A TEST RUN

23

variables to 2026-10-04, forecast days labelled; 17 of them every day from 1993

0.08 s

to open the Amphan cube from the disk cache (1.15 MB to the browser)

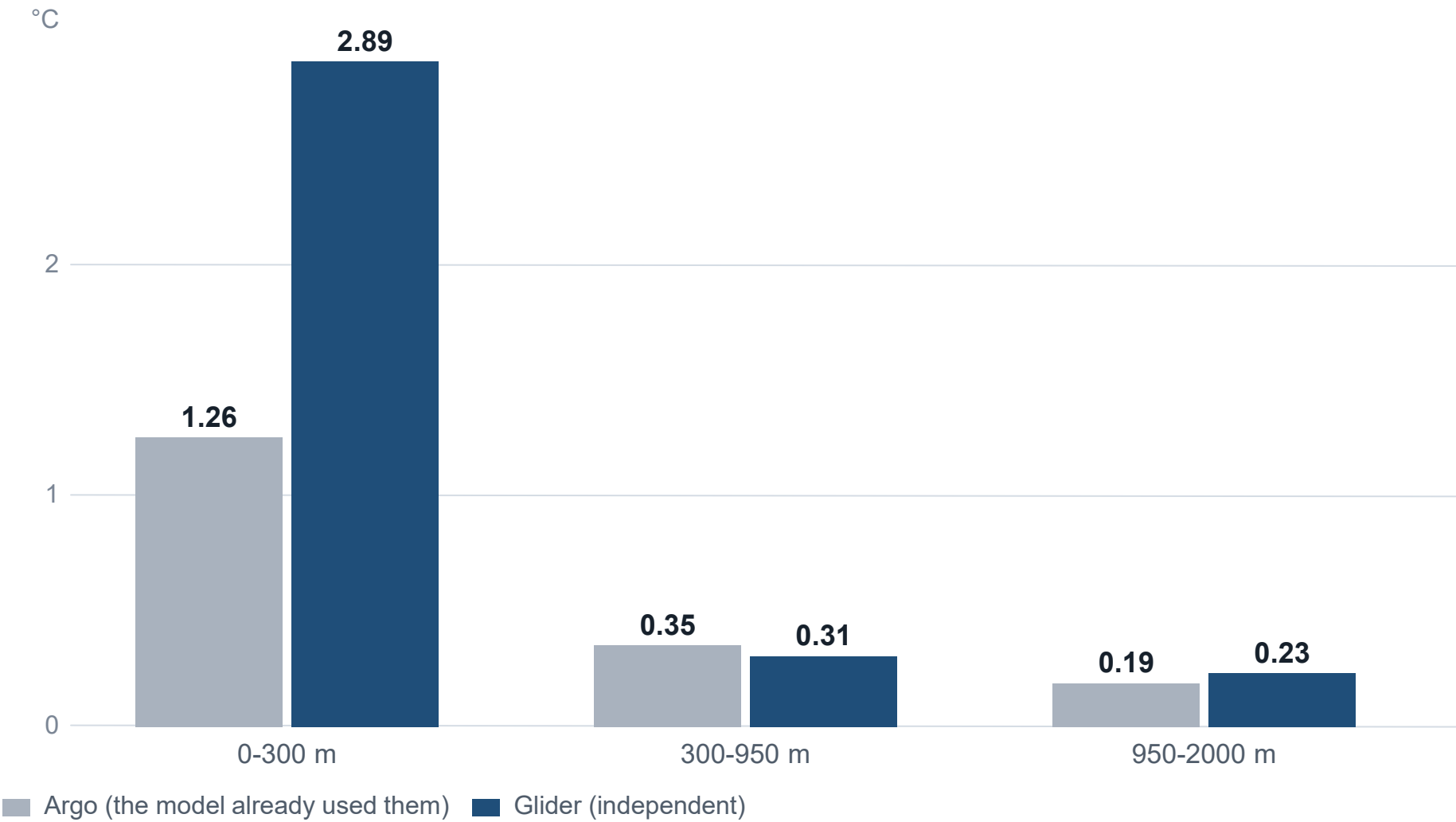
639 MB

peak memory over a full cold session: 97 of 97 requests answered

29 + 98

Argo profiles and glider casts co-located with the INCOIS model, level by level

WHAT THE PLATFORM FOUND: INCOIS MODEL ERROR BY DEPTH



Below 300 m the INCOIS model and the instruments agree. In the top 300 m, the warm layer a cyclone feeds on, the independent glider finds an error of **2.89 °C**. A single pooled number hides it; the 3D view shows it.

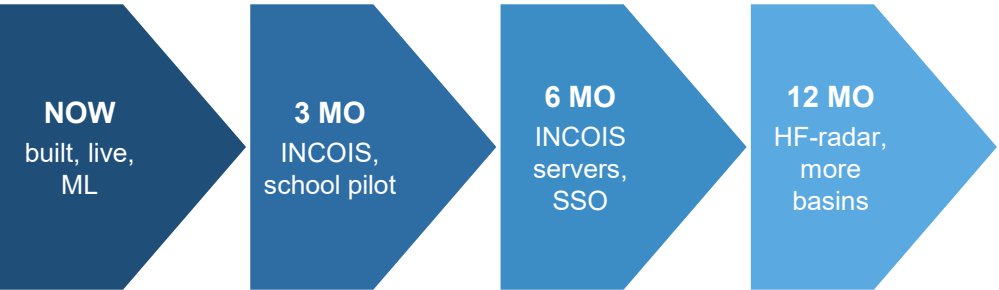
CHALLENGES AND RISKS, AND OUR STRATEGIES

CHALLENGE OR RISK	STRATEGY, ALREADY IN THE BUILD
The problem statement's data links are dead	Both are ftp:// and blocked; replaced with tested HTTPS sources.
The model fails QC too	6 INCOIS cells read above 40 °C at depth; masked, counted in the provenance panel.
Bad float readings	510 of 11,525 levels fail QC in the Bay; drawn red, never dropped.
An assistant could invent numbers	It reaches data only through our API; an untraceable number is flagged on screen.
A small host ran out of memory	Shared S3 client and capped caches; we ask for 1 GB, measured 639 MB peak.

SUSTAINABILITY

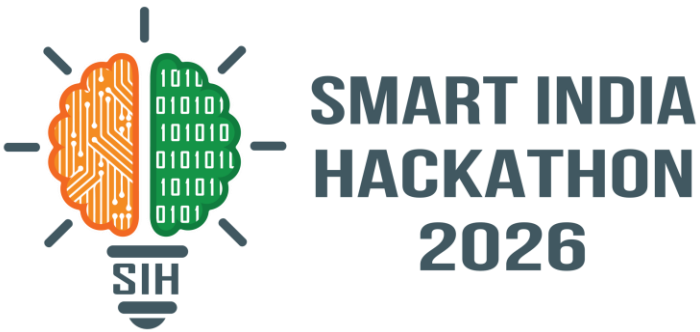
Free public data; only AI calls cost
Changed source = one fetch function
Test harness re-reads every source
Open source; issues and PRs open

ROADMAP



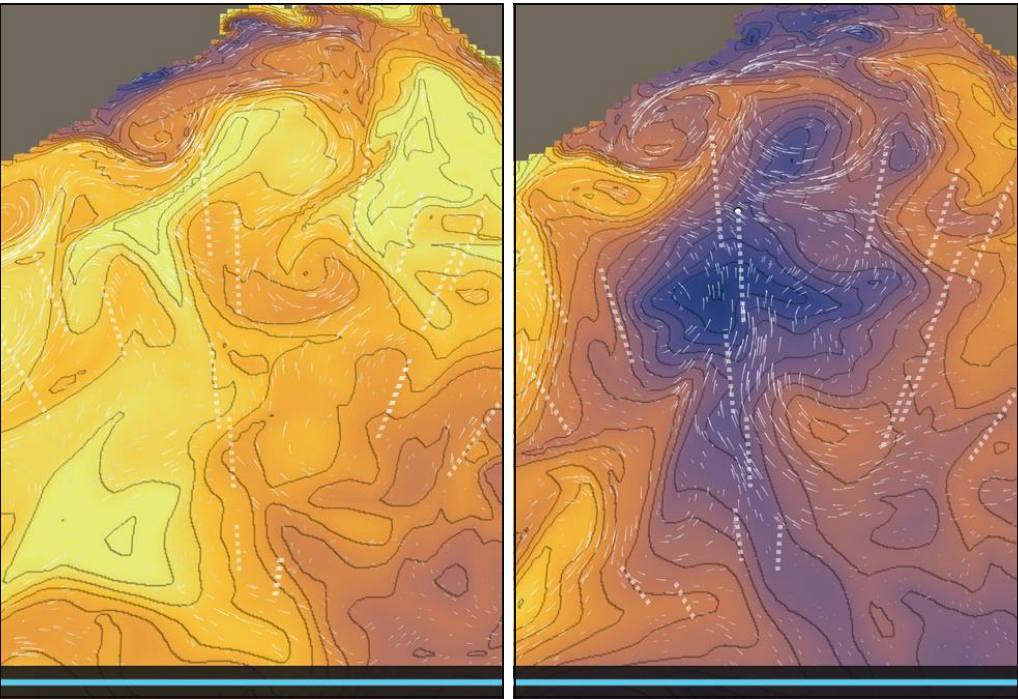


IMPACT AND BENEFITS



EACH TYPE OF USER AND THE IMPACT ON THEM

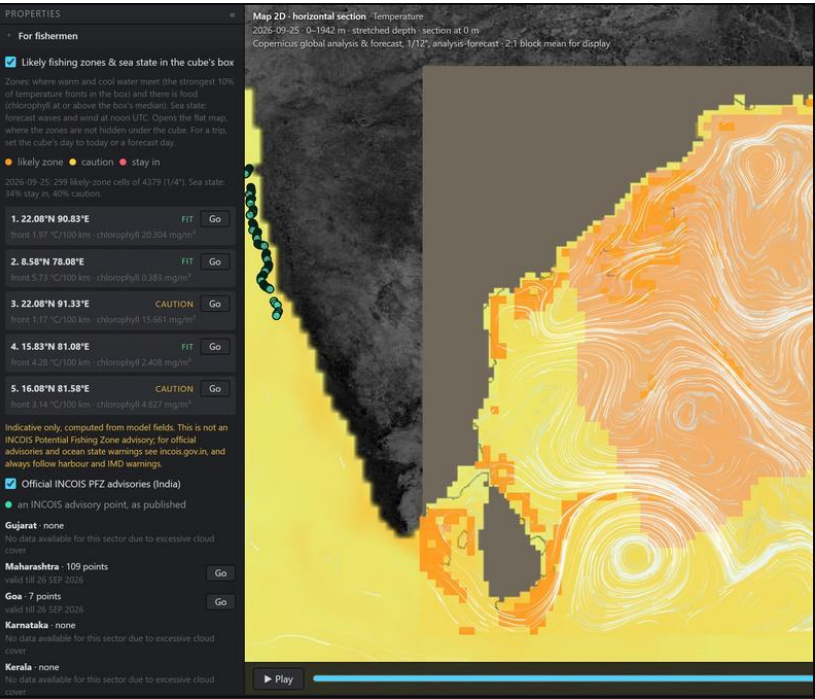
INCOIS forecasters



2020-05-14 2020-05-22

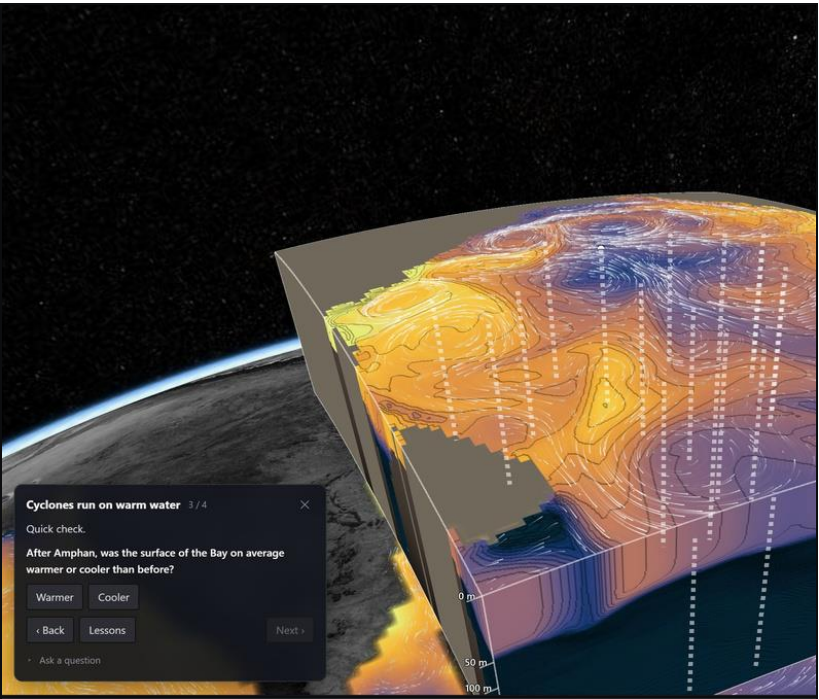
Before and after Cyclone Amphan, one 28–31.5 °C bar: the cold wake shows; the box's mean surface temperature is 0.77 °C lower after.

Fishermen



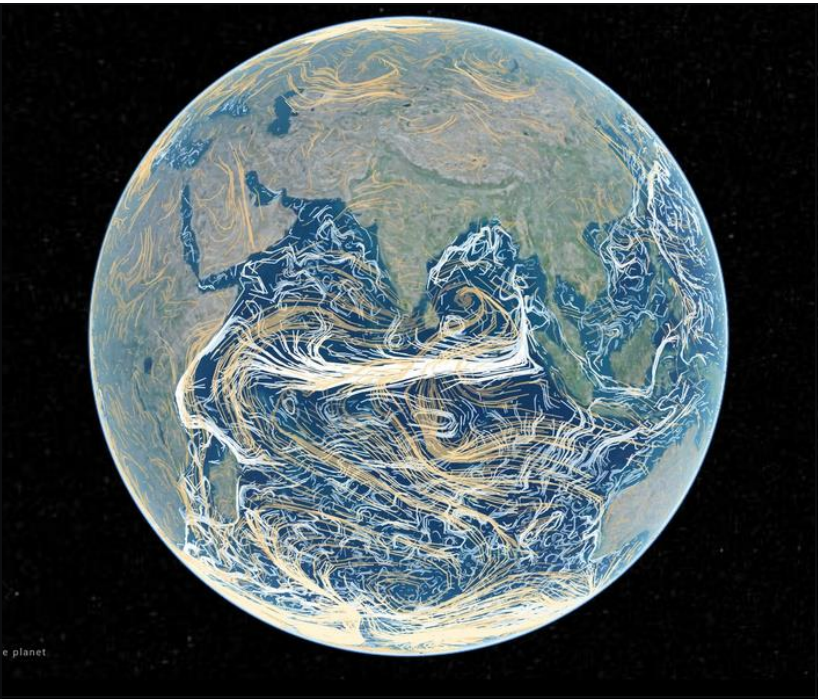
INCOIS's own fishing-zone advisories, all fourteen sectors as published, beside today's waves and wind. Our indicator says "not an advisory".

Students



Six lessons for class 8–12 on real data; quiz answers are read from the cube on screen, never from the language model.

Public and policymakers



Immersive view and cinematic tour: one day of the planet's currents and winds, for exhibitions and awareness campaigns.

BENEFITS

Scale: any box on Earth, not only the Bay; 6 other regions measured, under 0.4 s cached. Next: marine heatwaves, oil-spill drift.

Social

- Disaster preparedness: the warm layer a cyclone feeds on, seen in 3D, with cyclone heat potential per float
- Safety at sea: the official advisory first, with the sea state beside it
- Ocean literacy: a course in any school browser, on the same data forecasters use

Economic

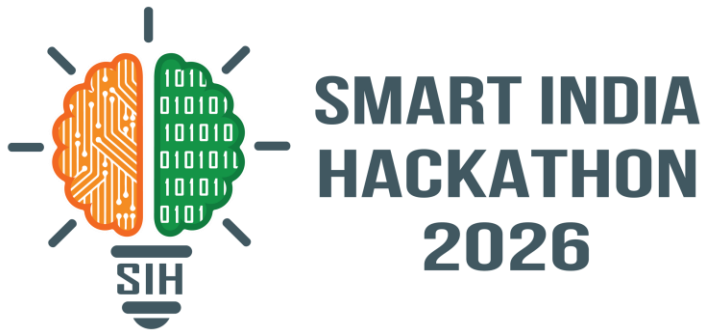
- Open source, no licences; a static site and one server (1 GB of memory measured enough)
- No archive to buy or store: public data is read where it lives, a day at a time
- Fishing advisories exist to cut search time at sea; here they sit where boats plan

Environmental

- Climate monitoring: every day since 1993 for 17 variables, in one view
- Oxygen-poor layers, chlorophyll and nutrients as 3D blocks, not flat maps
- Where the model is uncertain or unmeasured is shown, so decisions know the limits



RESEARCH AND REFERENCES



Source code, docs and every measured number: <https://github.com/Kukyos/VVater>

DATA, EACH SOURCE PROBED

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Copernicus Marine GLORYS12 and global analysis & forecast, physics and biogeochemistry, as ARCO stores. The cube.
data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030

2

INCOIS ERDDAP, Argo 10-day variational analysis with per-cell error. The Bay volume.
erddap.incois.gov.in/erddap

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Ifremer Argo ERDDAP, core and synthetic BGC floats. Floats anywhere.
erddap.ifremer.fr/erddap

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Argo GDAC, HTTPS mirror. Float files, QC flags, data mode.
data-argo.ifremer.fr

5

U.S. IOOS Glider DAC, Rutgers *ru29*. The independent check.
gliders.ioos.us/erddap

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INCOIS Potential Fishing Zone advisories. Read as published.
incois.gov.in/MarineFisheries/TextDataHome?mfid=1

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NCEP GFS 10 m wind via UCAR THREDDS. Wind forecast.
thredds.ucar.edu

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NASA GIBS Blue Marble relief. Land.
gibs.earthdata.nasa.gov

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Copernicus Marine In Situ TAC and MULTIOBS 3D BGC. Moorings, CTD, ADCP, HF-radar; the ML products.
data.marine.copernicus.eu/product/INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030

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doi.org/10.13155/33951

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U.S. IOOS QARTOD manuals. Glider QC.
ioos.noaa.gov/project/qartod

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CF Conventions for NetCDF. Read defensively.
cfconventions.org

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OGC Web Map Service 1.3.0. The WMS endpoint.
ogc.org/standards/wms

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IOC, SCOR, IAPSO (2010), TEOS-10. Density, sound speed.
teos-10.org

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Price (1981), J. Phys. Oceanogr. 11, 153–175. A cyclone cools the sea.

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India Meteorological Department, report on Super Cyclonic Storm Amphan (2020). Landfall, 20 May.

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WMO Guide to Wave Analysis and Forecasting (WMO-No. 702). Wind makes waves.

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Argo Program, "How Argo floats work". Floats.
argo.ucsd.edu

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SIH 2026 problem statement 26067, quoted verbatim.
github.com/Kukyos/VVater/blob/main/docs/01-problem-statement.md